CONFIDENTIAL MEMORANDUM

Rhino Bank Strategic Commodities Division

Yorkshire Sand Export Initiative

Developing a Regional Construction Materials Export Industry

Prepared for: Investment Committee, Bank of Leeds
Purpose: Strategic investment proposal
Sector: Natural Resources / Maritime Logistics / Construction Materials

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Executive Summary

This memorandum proposes the creation of **Yorkshire Sand & Aggregates Export PLC**, a Rhino Bank-backed company designed to transform Yorkshire's geological resources into an international export business.

The proposal is based on a simple economic opportunity:

> The Middle East has enormous quantities of desert sand, but often lacks the specific types of industrial and construction sand required for modern infrastructure.

Despite being surrounded by deserts, Gulf countries import construction-grade sand because wind-blown desert sand is often unsuitable for concrete due to its smooth particle shape and poor binding characteristics.

At the same time, global demand for construction sand is increasing rapidly, with the UN warning that demand for sand is rising faster than sustainable supply.

Yorkshire possesses:

- quarrying expertise
- industrial infrastructure
- Humber port access
- established aggregate supply chains
- high-quality processed sands and aggregates

Rhino Bank would finance a new export industry:

"Yorkshire Sand to the World"

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1. Strategic Opportunity

The Global Sand Market

Sand is one of the most consumed natural resources globally.

It is essential for:

- concrete
- glass
- construction
- infrastructure
- land reclamation
- specialist industrial uses

Global urbanisation is increasing demand, particularly in:

- Saudi Arabia
- UAE
- Qatar
- Bahrain
- Kuwait
- Oman

The Gulf region is undergoing major construction programmes, including:

- new cities
- airports
- infrastructure
- industrial zones
- tourism developments

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2. Why Export Yorkshire Sand?

The misconception is:

> "The Middle East has deserts, therefore it has unlimited construction sand."

However, construction requires specific grain properties.

Desert sand is often too rounded and fine for high-strength concrete, meaning imported sand and aggregates are required.

Yorkshire can supply:

- washed sand
- sharp sand
- silica sand
- specialist construction aggregates

Yorkshire already has an established aggregates industry producing construction sands and related materials.

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3. Proposed Company

Yorkshire Sand & Aggregates Export PLC

A Rhino Bank-backed natural resources company.

Mission:

> Convert Yorkshire's geological assets into export revenues while creating jobs and industrial capacity.

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4. Operating Model

Supply Chain

Step 1

Sand extracted from approved Yorkshire sources.

Potential suppliers:

- existing quarries
- new permitted extraction sites
- recycled aggregate facilities

Step 2

Processing facilities:

Located near:

- Leeds
- Sheffield
- Hull
- Humber ports

Processing:

- washing
- grading
- drying
- quality certification

Step 3

Export through:

Port of Hull / Humber Estuary

Advantages:

- deep-water access
- industrial infrastructure
- proximity to Yorkshire resources
- shipping connections

Step 4

Bulk shipping

Similar principle to the water proposal:

Utilise:

- bulk carriers
- aggregate vessels
- return maritime capacity

Potential destinations:

- Dubai
- Abu Dhabi
- Doha
- Riyadh via Gulf ports

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5. Initial Investment Proposal

Yorkshire Sand Export Terminal

Capital Requirement

Asset	Cost
Processing plant	£15m
Storage facilities	£5m
Port loading infrastructure	£10m
Quarry investment	£10m
Machinery	£5m
Working capital	£5m

Total Investment:

£50 million

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6. Bank of Leeds Funding Structure

Yorkshire Strategic Resources Fund

Investor	Amount
Bank of Leeds	£20m
Yorkshire pension funds	£10m
Private investors	£10m
Industry partners	£10m

Total:

£50 million

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7. Export Economics

Conservative Pilot Scenario

Annual exports:

500,000 tonnes

Equivalent:

approximately 20–25 large bulk shipments annually.

Selling Price

Assume:

Export price:

£35 per tonne

Revenue:

500,000 × £35

£17.5 million annually

Operating Costs

Cost	Annual
Extraction	£3m
Processing	£3m
Port handling	£2m
Shipping contribution	£4m
Administration	£1m

Total:

£13 million

Operating Profit

Revenue:

£17.5m

Costs:

£13m

EBITDA:

£4.5 million

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8. Expansion Scenario

Regional Export Scale

Increase exports:

2 million tonnes annually

Revenue:

2,000,000 × £35

£70 million

Costs:

≈ £45 million

EBITDA:

£25 million

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9. Investment Return

At mature scale:

EBITDA:

£25m

Industry valuation:

6× EBITDA

Potential company value:

£150 million

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10. Employment Impact

Direct Jobs

Area	Jobs
Quarry operations	80
Processing	60
Logistics	40
Engineering	20
Administration	20

Direct:

220 jobs

Indirect:

500+ jobs

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11. Yorkshire Industrial Benefits

The project strengthens:

Mining and Quarrying

Revitalising Yorkshire's historic resource economy.

Manufacturing

Demand for:

- machinery
- engineering
- automation

Ports

Increasing Humber maritime activity.

Logistics

Creating export supply chains.

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12. Strategic Expansion Opportunities

The company could expand into:

Specialist Materials Division

Products:

- silica sand
- glass-making sand
- industrial minerals
- filtration sands
- sports sands

Higher-value materials could command significantly higher margins than basic construction sand.

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13. Environmental Strategy

The Bank of Leeds would require:

- regulated extraction
- restoration of quarry sites
- biodiversity improvements
- recycled aggregate integration
- transparent environmental reporting

The objective is not uncontrolled extraction, but responsible resource development.

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14. Risks

Environmental Regulation

Mitigation:

Operate only with approved extraction plans.

Shipping Costs

Mitigation:

Use bulk shipping contracts and return capacity.

Competition

Mitigation:

Focus on premium Yorkshire-certified materials.

Commodity Prices

Mitigation:

Long-term supply contracts with Gulf construction firms.

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15. Strategic Recommendation

The Bank of Leeds should establish:

Yorkshire Strategic Resources Fund

Initial portfolio:

1. Yorkshire Sand Export PLC
2. Yorkshire Water Export Initiative
3. Yorkshire Minerals Investment
4. Industrial recycling businesses

Final Investment Summary

Metric	Estimate
Initial investment	£50m
Export capacity	500k–2m tonnes/year
Mature revenue	£70m/year
Mature EBITDA	£25m/year
Potential valuation	£150m
Direct jobs	220
Indirect jobs	500+

Conclusion

Yorkshire has historically built its wealth through:

- coal
- steel
- engineering
- manufacturing

The next generation of industrial strategy should identify new global markets.

A Bank of Leeds-backed sand export industry would transform a basic geological resource into:

- export income
- industrial employment
- port activity
- regional investment returns

The proposal represents a modern version of Yorkshire's historic resource economy:

From the mines and quarries of the past, to the strategic resources economy of the future.